GC–MS RETENTION-TIME STANDARDISATION

Joint-Align: Intuition, Visuals & Mathematics

How shared latent clocks, piecewise hat bases, and bounded convex solvers guarantee path-consistent alignment.

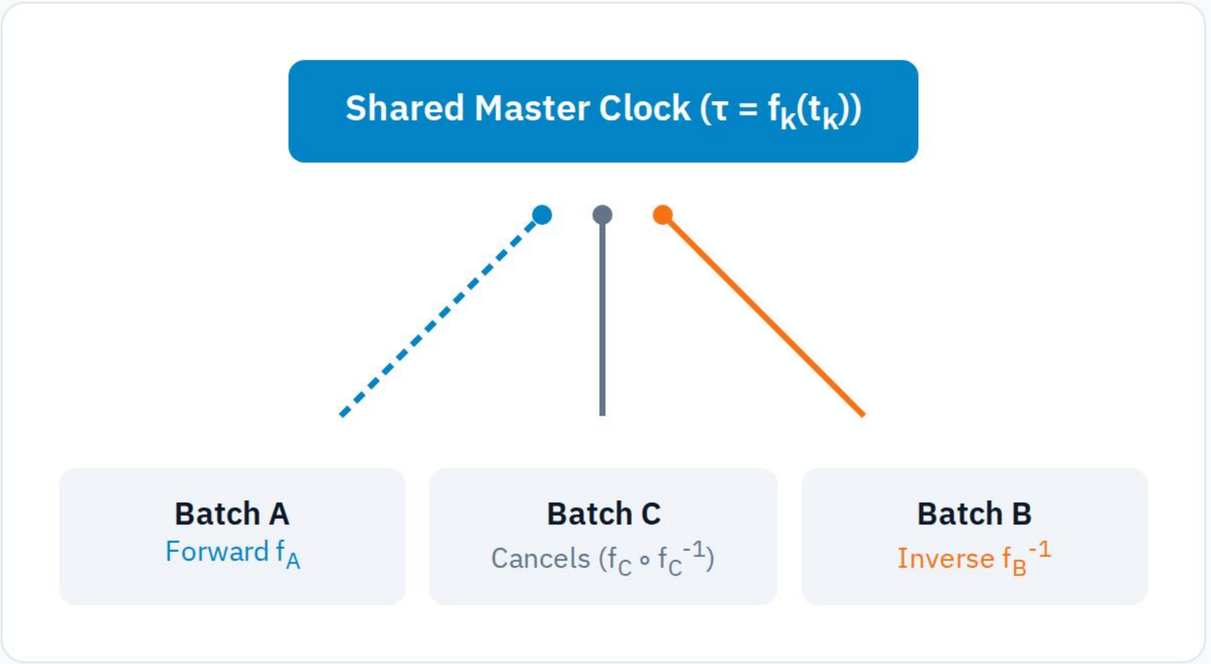
The Master Clock & Exact Path Transitivity

Pairwise Failure: Independent fits drift across batch triplets.

Shared Axis: Every batch maps forward into one master clock.

Telescoping Hops: Intermediate batch detours cancel via identity.

$$g_{A \to B} = f_B^{-1} \circ (f_C \circ f_C^{-1}) \circ f_A = f_B^{-1} \circ f_A$$



02 / EMPIRICAL VERIFICATION

Triplet Discrepancy: Pairwise vs. Joint Fit

0.116m

Mean Pairwise Triplet Error

0.277m

Max Pairwise Drift (Exceeds Bin)

0.0023m

Max Joint Discrepancy (Grid Rounding)

Transitivity is guaranteed by parameterisation; empirical accuracy still requires high-quality anchors.

Fitting Without Function Inverses

$$r_j(\theta) = f_k(a_j) - f_l(b_j) = (a_j - b_j) + B(a_j)L\theta_k - B(b_j)L\theta_l$$

During Optimization (Fitting)

Forward Projection: Both batches map to master clock.

Zero Inverses: No f_l^{-1} or inverse Jacobians needed.

Purely Linear: Residual is linear in parameter vector θ.

Downstream Alignment (Post-Fit)

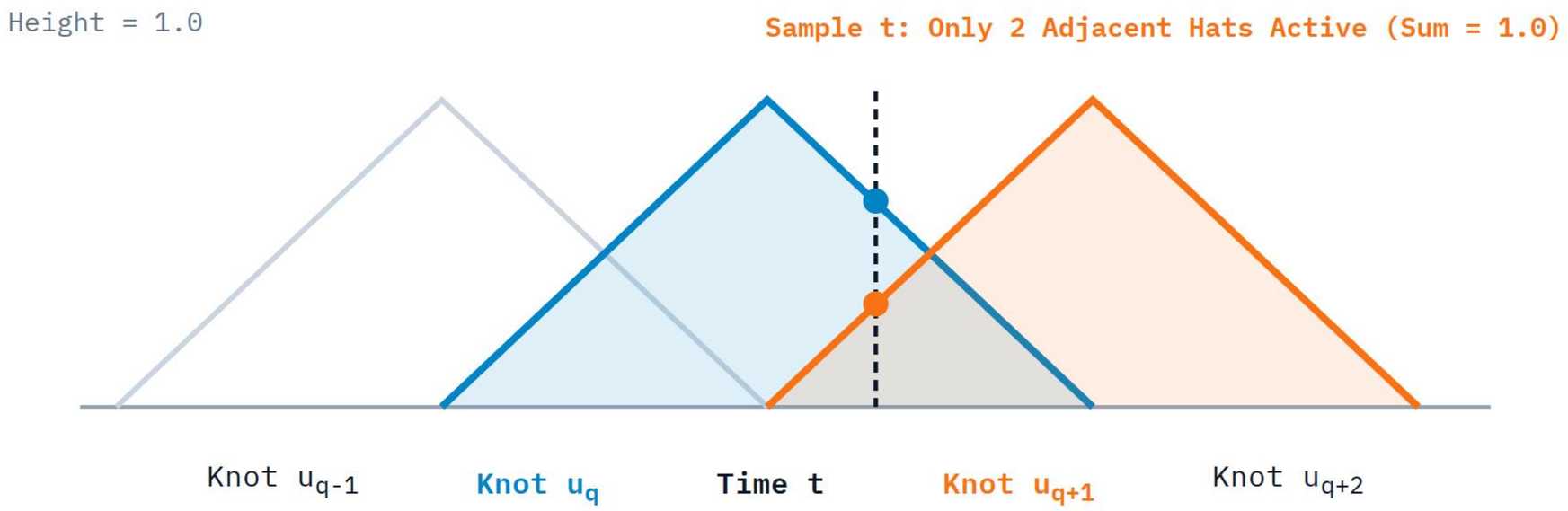
Signal Overlay: Evaluates composite map $f_l^{-1} \circ f_k$.

Guaranteed Inverse: Strict monotonicity ensures unique f_l^{-1}.

Fast Lookup: Computed via 1D grid interpolation.

04 / CONTINUOUS INTERPOLATION

Piecewise Linear Hat Basis B(t)



Fixed Geometry B(t): Pre-computed interpolation weights. **Learned Values Lθ_k:** Cumulative drift at discrete knots.

Converting Monotonicity into Box Constraints

Feature	Raw Knot Values (V_k)	Step Increments (θ_k with Matrix L)
Drift Formula	$d_k(t) = B(t)\, V_k$	**$d_k(t) = B(t)\, L\, \theta_k$**
Slope Bound [0.5, 2.0]	$-0.5\Delta u \le v_{q+1} - v_q \le 1.0\Delta u$	**$-0.5\Delta u \le \delta_{kq} \le 1.0\Delta u$**
Constraint Coupling	Coupled adjacent parameters	**Independent parameter box bounds**
Required Solver	General Quadratic Programming (QP)	**Bounded-Variable Least Squares (BVLS)**

$$\min_{\theta} \| A\theta - y \|_2^2 \quad \text{subject to} \quad lb \le \theta \le ub$$

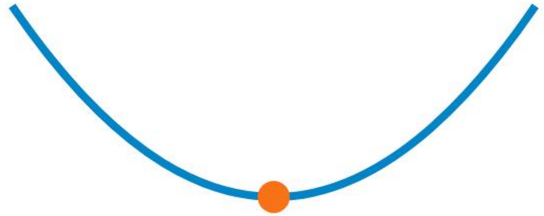
Resolving Gauge Freedom: Trough vs. Bowl

Without Gauge (Positive Semi-Definite)



Flat Valley ($\lambda_{min} = 0$): Shifting all clocks by +c leaves pairwise residuals unchanged.

With Gauge Penalty (Strictly Positive Definite)



Unique Global Minimum ($\lambda_i > 0$): Reference or Mean gauge pins the master clock.

07 / OUTLIER REJECTION

Two-Stage IRLS: Sample Re-Weighting

STEP 01

MAD Scale Estimate

Compute robust anchor residual scatter.

→

STEP 02

Huber Warm-Start

Convex L1/L2 basin limits extreme pulls.

→

STEP 03

Tukey Bisquare Cut

Redescending weight sets false anchors to 0.

→

STEP 04

Weighted BVLS

Scale rows by $\sqrt{w_j}$ until convergence.

Unlike Adam (which scales parameter learning rates), IRLS re-weights noisy data samples (anchors).

Why Sample-Wise Cross-Validation Fails

Compound Leakage

Identical peaks appear across remaining training samples.

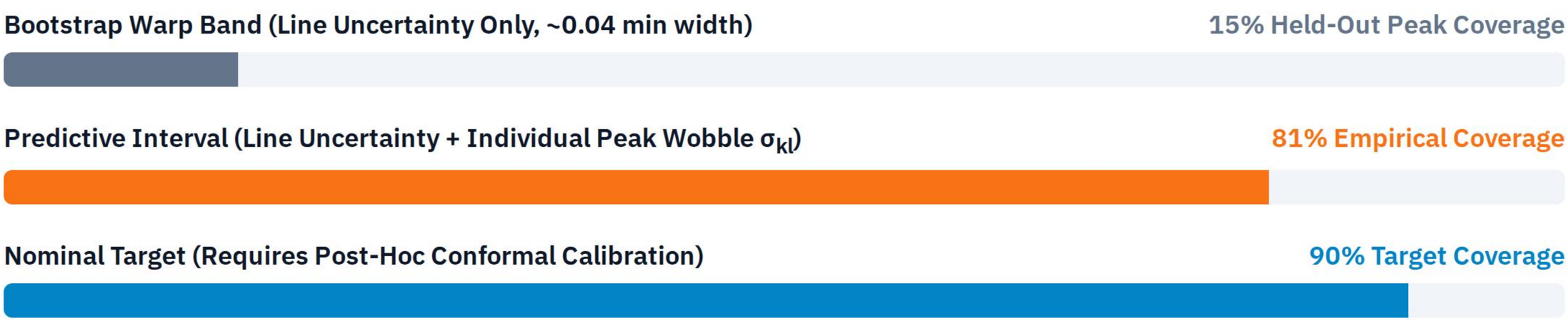
Hidden Gap Wiggles

Sample holdouts never test empty retention windows.

Contiguous RT Blocks

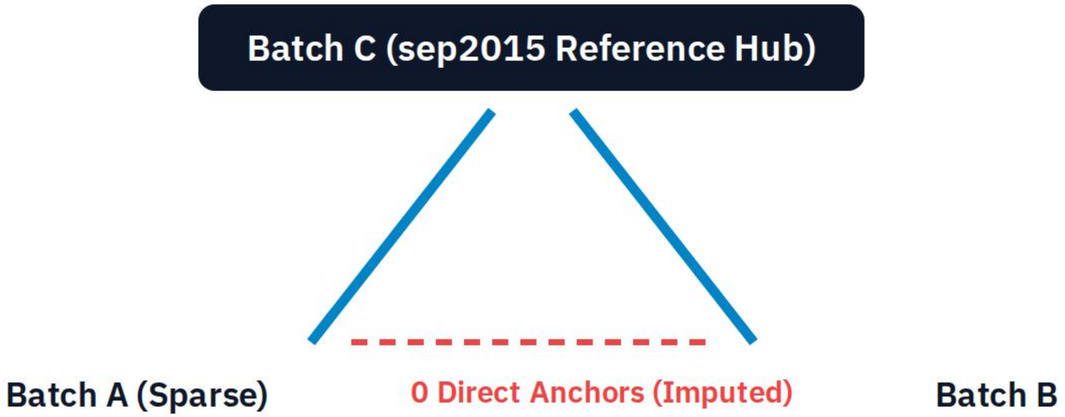
Holding out 2.7-minute windows forces gap interpolation.

Warp Band vs. Predictive Interval Coverage



Why 81% instead of 90%? Sample bootstrap captures sampling variance but misses smoothing bias from penalty λ.

Borrowing Strength Across Partial Constraints



Rigid Frame Analogy: Mutual links to Batch C lock relative positions.

Leave-Pair-Out Test: 100% of direct A–B anchors deleted.

Stable Recovery: Alignment changes by ≤ 0.007 TIC correlation across 9/10 pairs.

ARCHITECTURAL SUMMARY

One Unified, Convex & Transitive Linear System

Forward master-clock residuals + hat basis weights + cumulative step increments = fast, globally optimal multi-batch alignment.